Deep Learning for Audio

Lecture 6

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AI Masters

2025

Outline

1. Neural Vocoders
2. Parallel WaveGAN
3. Hi-Fi GAN
4. DiffWave
5. Flows
6. VITS

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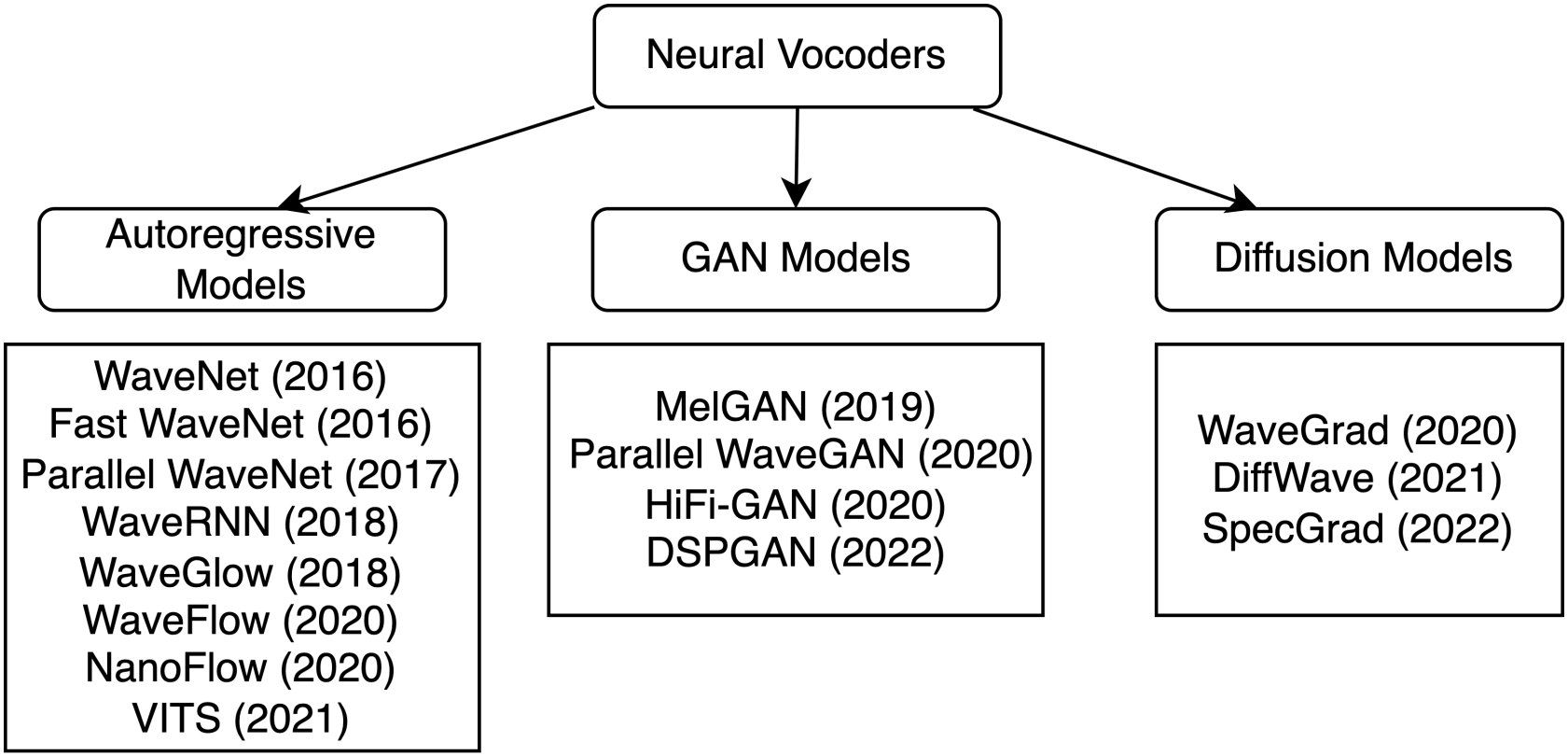
Neural Vocoders



Neural Vocoders

Model type	**Model**	**MOS on LJ Speech**
Autoregressive	WaveNet	3.68
	WaveRNN	3.96
GAN	MelGAN	3.73
	Parallel WaveGAN	3.99
Diffusion	WaveGrad	3.85
	DiffWave	**4.07**
	Griffin-Lim	3.68
	Ground Truth	**4.10**

AlBadawy et al., Vocbench: A Neural Vocoder Benchmark for Speech Synthesis, IEEE ICASSP, 2022

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Parallel WaveGAN

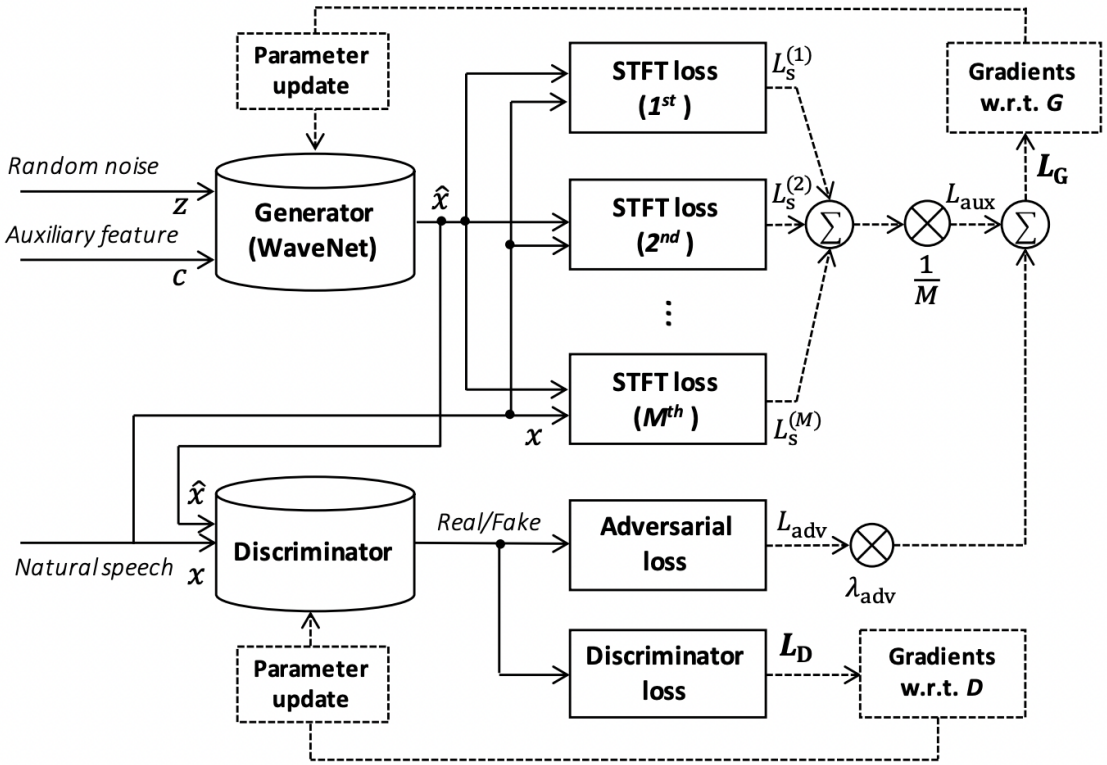
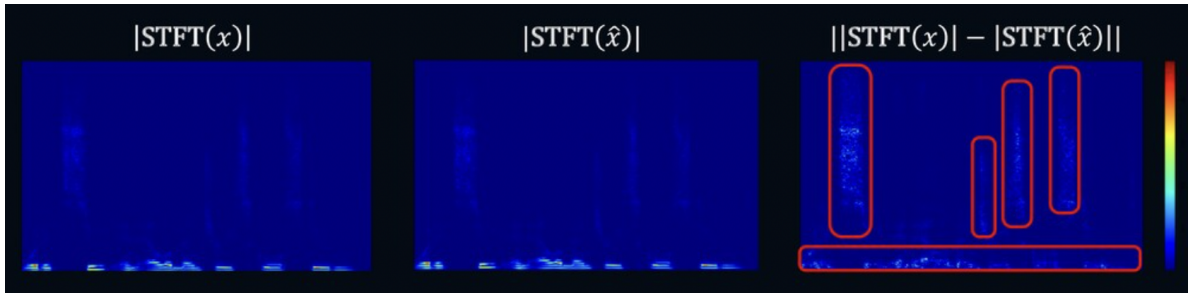


Figure: Parallel WaveGAN

Yamamoto et al., Parallel Wavegan: A Fast Waveform Generation Model Based on Generative Adversarial Networks with Multi-Resolution Spectrogram, IEEE ICASSP, 2020

STFT Loss



$$\mathcal{L}_{\mathrm{s}}(G) = \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z}),\mathbf{x}\sim p_{data}}\left[\mathcal{L}_{\mathrm{sc}}(\mathbf{x},\hat{\mathbf{x}}) + \mathcal{L}_{\mathrm{mag}}(\mathbf{x},\hat{\mathbf{x}})\right]$$

$$\mathcal{L}_{\mathrm{sc}}(\mathbf{x},\hat{\mathbf{x}}) = \frac{\left\| |\mathrm{STFT}(\mathbf{x})| - \left|\mathrm{STFT}^{\mathrm{T}}(\hat{\mathbf{x}})\right| \right\|}{\left\| |STFT(\mathbf{x})| \right\|_F}$$

$$\mathcal{L}_{\mathrm{mag}}(\mathbf{x},\hat{\mathbf{x}}) = \frac{1}{N}\left\| \log|\mathrm{STFT}(\mathbf{x})| - \log|\mathrm{STFT}(\hat{\mathbf{x}})| \right\|_1$$

$$\mathcal{L}_{\mathrm{aux}}(G) = \frac{1}{M}\sum_{m=1}^{M}\mathcal{L}_{\mathrm{s}}^{(m)}(G), \text{M – number of STFT losses}$$

Takaki et al., STFT Spectral Loss for Training a Neural Speech Waveform Model, IEEE ICASSP, 2019

GAN Loss

- Discriminator Loss:

$$\mathcal{L}_{\mathrm{D}}(G, D) = \mathbb{E}_{\mathbf{x} \sim p_{\mathrm{data}}}[(1 - D(\mathbf{x}))^2] + \mathbb{E}_{\mathbf{z} \sim N(0,I)}[D(G(\mathbf{z}))^2]$$

- Generator Loss

$$L_{\mathrm{adv}}(G, D) = \mathbb{E}_{\mathbf{z} \sim N(0,I)} \left[(1 - D(G(\mathbf{z})))^2\right]$$

$$L_{\mathrm{G}}(G, D) = L_{\mathrm{aux}}(G) + \lambda_{\mathrm{adv}} L_{\mathrm{adv}}(G, D)$$

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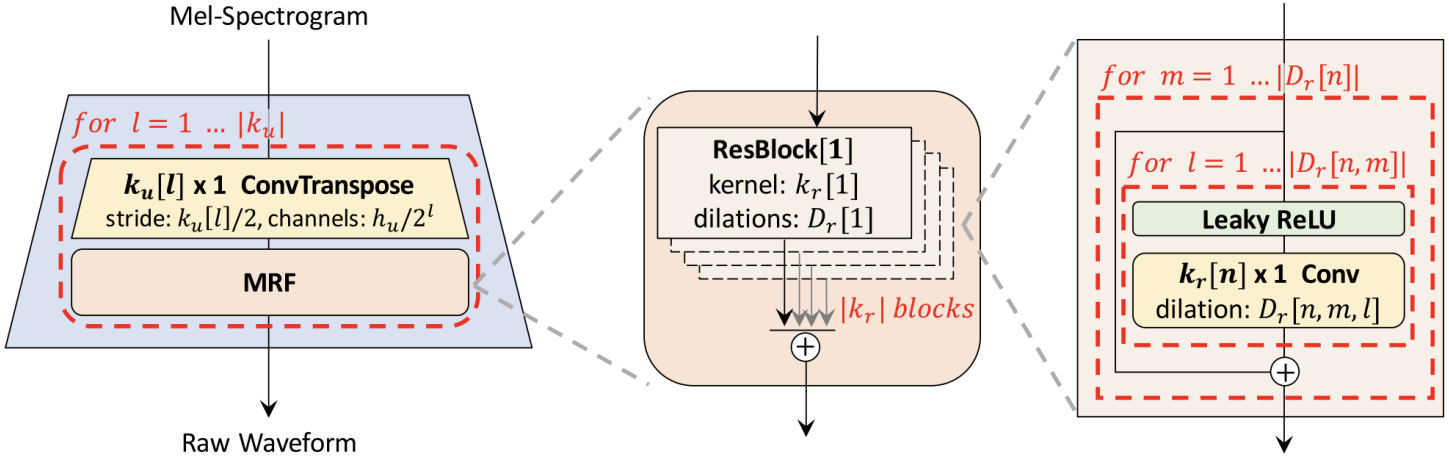
Hi-Fi GAN: generator

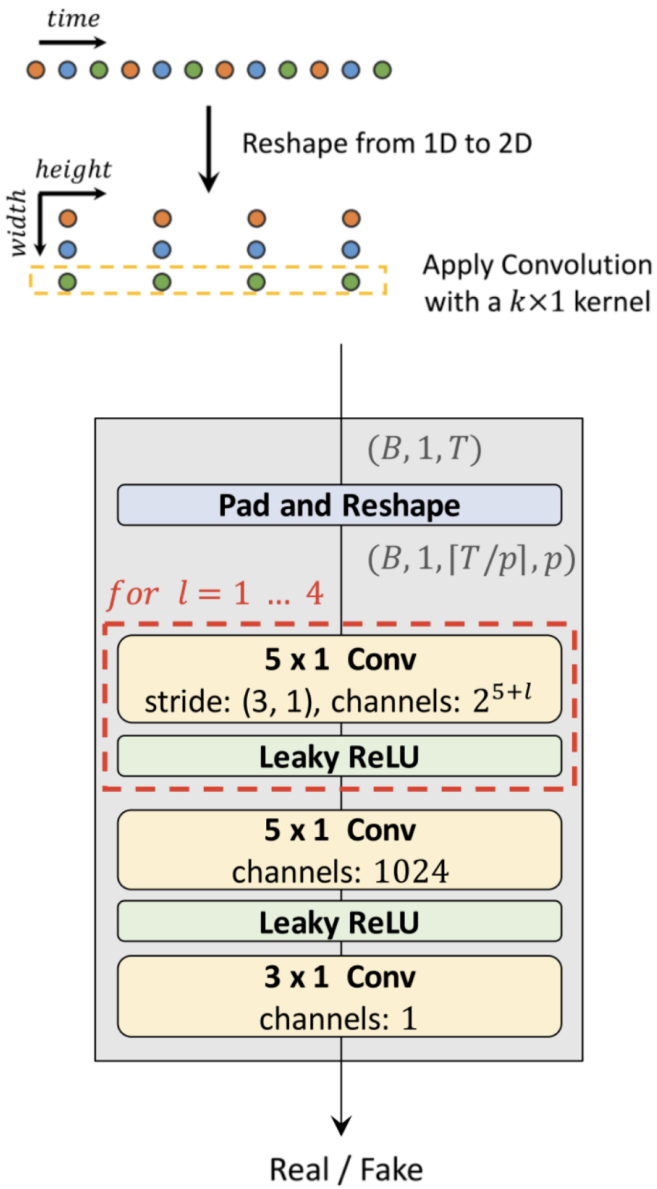


- Fully-Convolutional Architecture
- Upsamples mel-spectrogram to raw audio via transposed convolutions
- **Multi-Receptive Field (MRF)** Fusion observes patterns of various lengths in parallel and aggregates the outputs from multiple residual blocks.

Kong et al., HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis, NeurIPS, 2020

Hi-Fi GAN: Multi-Period Discriminator (MDP)

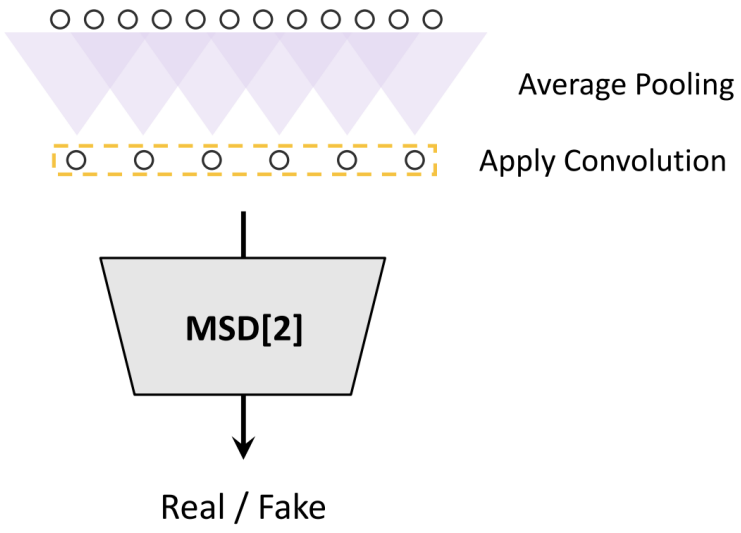
- MPD: mixture of sub-discriminators
- Sub-discriminator: stack of strided conv layers
- Sub-discriminator only accepts equally spaced samples of an input audio
- The space is given as period p = [2, 3, 5, 7, 11] (defined by the authors to minimize overlaps)



Hi-Fi GAN blogpost

Hi-Fi GAN: Multi-Scale Discriminator (MSD)

- Sub-discriminator in MPD only accepts disjoint samples $\Rightarrow$ add MSD to consecutively evaluate the audio sequence
- MSD: three sub-discriminators working on different scales: raw audio, 2-x avg-pooled audio, 4-x avg-pooled audio.
- Each sub-discriminator in MSD: stacked conv layers with leaky ReLU activation



Hi-Fi GAN: losses

$$\mathcal{L}_{adv}^{D} = \mathbb{E}_{(x,c)}[(D(x)-1)^2 + D(G(c))^2]$$

$$\mathcal{L}_{adv}^{G} = \mathbb{E}_{c}[(D(G(c))-1)^2]$$

$$\mathcal{L}_{mel}(G) = \mathbb{E}_{(x,c)}[||\phi(x) - \phi(G(c))||_1]$$

$$\mathcal{L}_{feat}(G,D) = \mathbb{E}_{(x,c)}\left[\sum_{l=1}^{L} ||D^{(l)}(x) - D^{(l)}(G(c))||_1\right]$$

- x – ground truth
- $s = \phi(x)$ – mel-spectrogram
- D is a single discriminator from the ensemble

Hi-Fi GAN: overall loss

$$\mathcal{L}_G = \sum_{k=1}^{K} \left[\mathcal{L}_{adv}^{D_k} + \lambda_{feat} \mathcal{L}_{feat}(G, D_k) \right] + \lambda_{mel} \mathcal{L}_{mel}(G)$$

$$\mathcal{L}_D = \sum_{k=1}^{K} \mathcal{L}_{adv}^{D_k}$$

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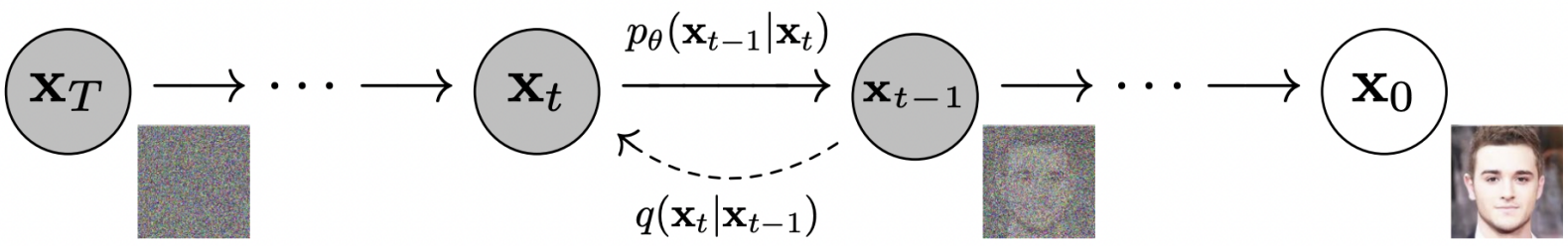
Diffusion models idea



- **Diffusion probabilistic model**: parameterized Markov chain from data x_0 to the latent variable x_T

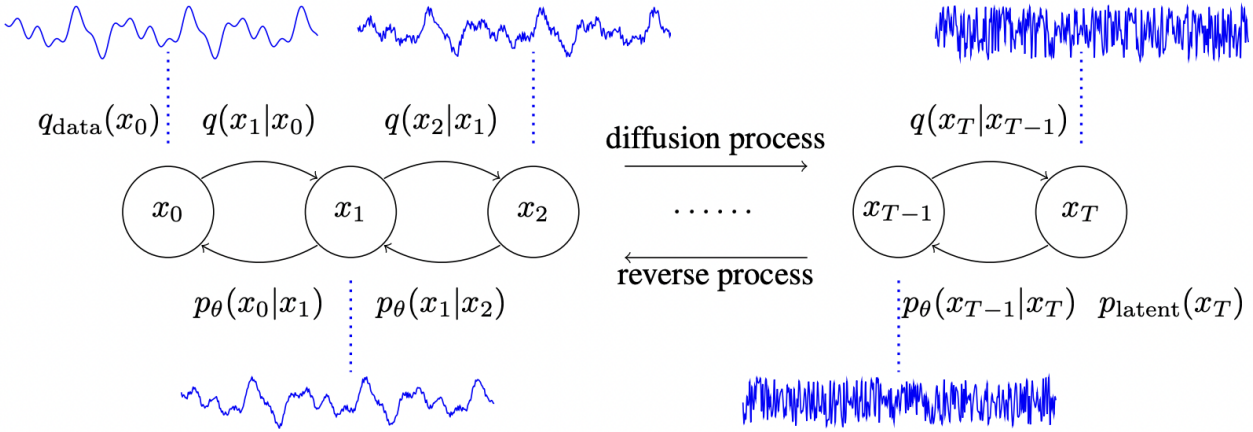
$$q(x_1, \cdot \cdot \cdot \cdot \cdot x_T | x_0) = \prod_{t=1}^{T} q(x_t | x_{t-1})$$

- **Reverse process**: Markov chain from x_T to x_0 parameterized by θ:

$$p_{\text{latent}}(x_T) = \mathcal{N}(0, I), \quad p_\theta(x_0, \cdots, x_{T-1} | x_T) = \prod_{t=1}^{T} p_\theta(x_{t-1} | x_t)$$

Ho et al., Denoising Diffusion Probabilistic Models, NeurIPS Proceedings, 2020

DiffWave



- **Sampling**: reverse process $x_T \sim \mathcal{N}(0, I)$, $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$ for $t = T, T-1, \cdots, 1$. x_0 – sampled data.
- **Training**: $p_\theta(x_0) = \int p_\theta(x_0, \cdots, x_{T-1}|x_T) \cdot p_{\text{latent}}(x_T) \mathbf{d}x_{1;T}$ (likelihood) is intractable to calculate $\Rightarrow$ model trained by maximizing its variational lower bound (ELBO):

$$\mathbb{E}_{q_{\text{data}}(x_0)} \log p_\theta(x_0) \geq$$
$$\geq \mathbb{E}_{q(x_0,\cdots,x_T)} \log \frac{p_\theta(x_0, \cdots, x_{T-1}|x_T) \cdot p_{\text{latent}}(x_T)}{q(x_1, \cdots, x_T|x_0)} := \text{ELBO}$$

Kong et al., DiffWave: A Versatile Diffusion Model for Audio Synthesis, ICLR, 2021

DiffWave: advantages

- **Non-autoregressive**: much faster than WaveNet
- **Compact model**: smaller footprint than flow-based models
- No auxiliary losses in training (e.g., spectrogram-based losses): **no mode collapse** like in GANs/VAEs

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Flow-based models basic idea

- Use an invertible transformation $z = f_\theta(x)$ to transform x into z;
- Suppose that z has a simple distribution $p(z)$ (e.g. standard normal);
- Use maximum likelihood to fit θ.

$$\sum_{i=1}^{n} \log p_\theta(x_i) = \sum_{i=1}^{n} \log p(z_i) + \log \left| \det \left(\frac{dz_i}{dx_i} \right) \right| \xrightarrow{\max}_\theta$$

- $p(z)$ is easy to compute. But what about log-abs-det-Jacobian?

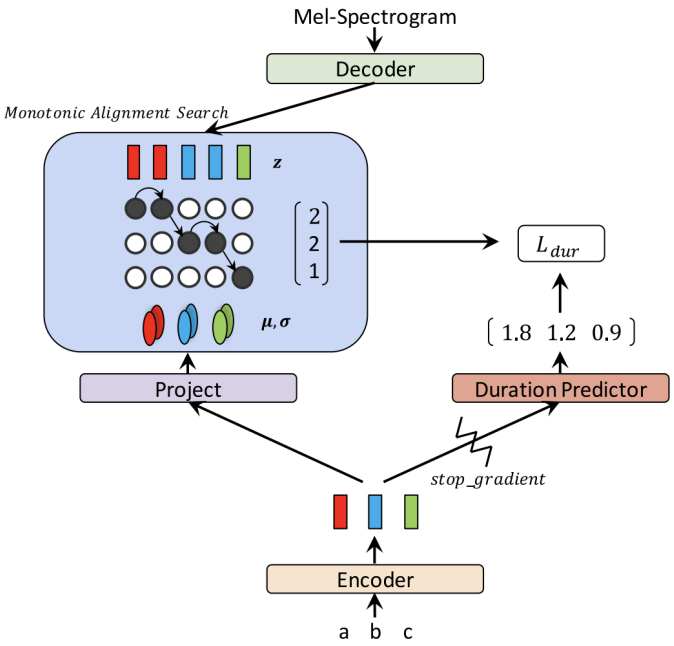
Flow-based models

- We suppose that $f_\theta = f^{(L)} \circ f^{(L-1)} \circ \cdots \circ f^{(1)}$.
- If each of $f^{(j)}$ is invertible, then

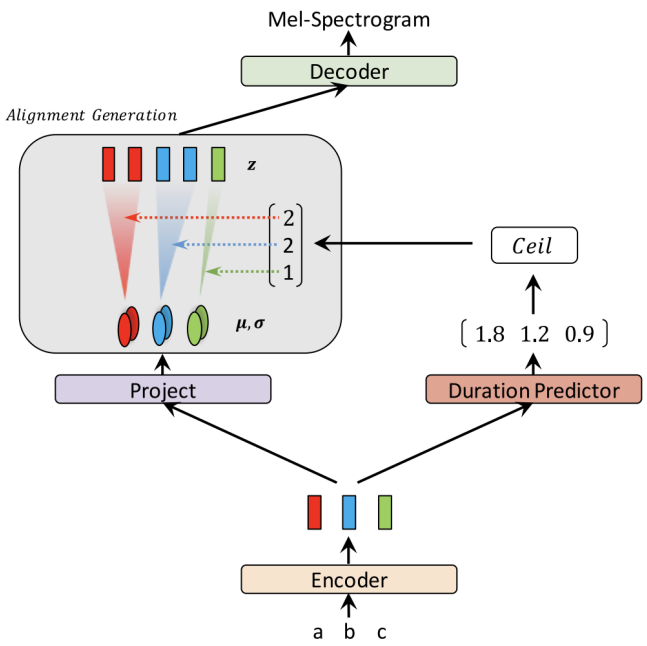
$$\log \left| \det \left(\frac{dz}{dx} \right) \right| = \sum_{j=1}^{L} \log \left| \det \left(\frac{dh_j}{dh_{j-1}} \right) \right|,$$

  where $h_j = f_j(x)$, $h_0 = x$, and $h_L = z$.
- We now have to come up with invertible parametric transformations such that they have easily computable log-abs-det-Jacobian and their composition is expressive enough.
- The resulting transformation f is called a *normalizing flow* or just *flow*.
- We can sample from the model just by applying the inverse transform $x = f_\theta^{-1}(z)$:

Glow-TTS



(a) An abstract diagram of training procedure.



(b) An abstract diagram of inference procedure.

- Latent space leads to more diversity
- More natural spectrograms
- Not autoregressive

Kim et al., Glow-TTS: A Generative Flow for Text-to-Speech via Monotonic Alignment Search, NeurIPS, 2020

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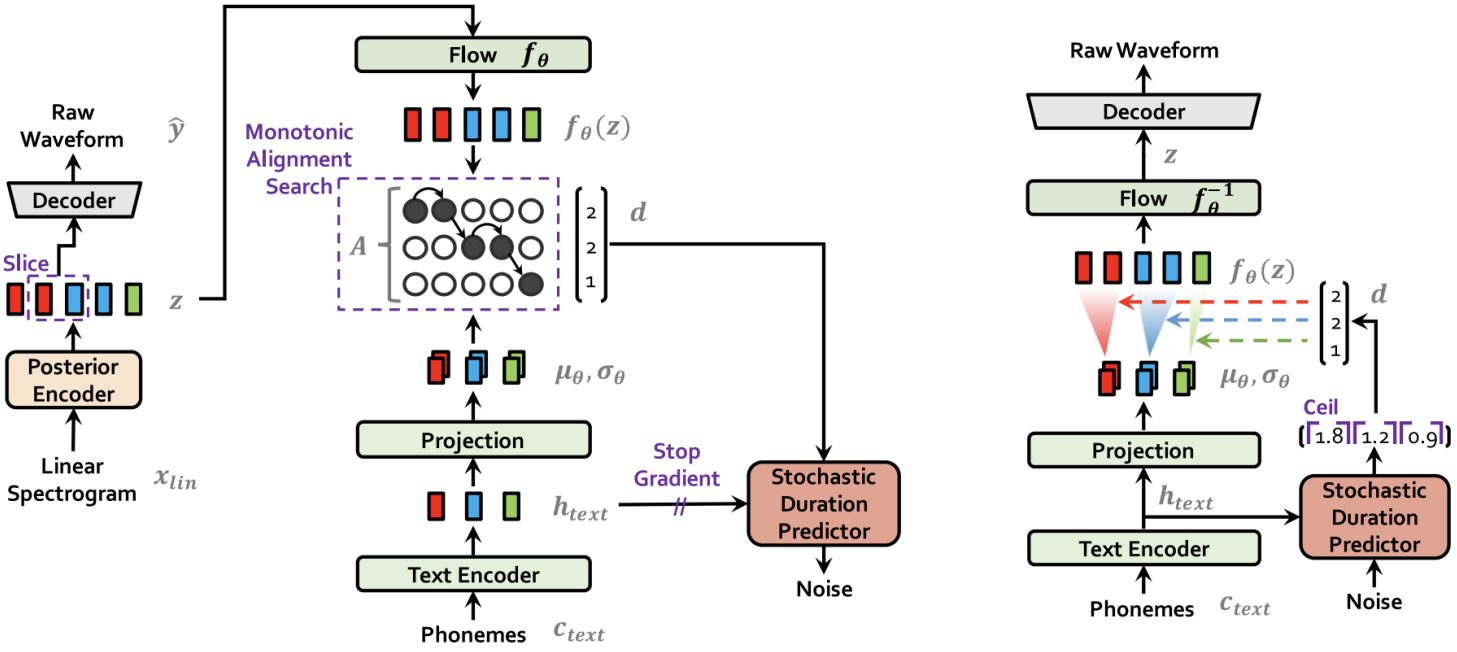
VITS (**V**ariational **I**nference with adversarial learning for end-to-end **T**ext-to-**S**peech)



(a) Training procedure

(b) Inference procedure

- Conditional VAE with the objective of maximizing ELBO (conditional prior: green blocks)
- **Inference**: text ⇒ duration prediction ⇒ cascade of the flow module and HiFi-GAN decoder ⇒ waveform

Kim, Jaehyeon et al., Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech, ICML, 2021

VITS blocks

- **Posterior encoder**: non-causal WaveNet residual blocks
- **Prior encoder**: text encoder that processes the input phonemes c_{text} and a normalizing flow f_θ that improves the flexibility of the prior distribution
- **Text encoder**: transformer encoder with relative positional representation
- **Stochastic duration predictor**: flow-based generative model
- **Normalizing flow**: stack of affine coupling layers consisting of a stack of WaveNet residual blocks
- **Decoder**: HiFi-GAN V1 generator